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The importance of the individual predictors and the psychosocial working conditions in assessing the Work Ability Index of people with low vision

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The importance of the individual predictors and the psychosocial working conditions in assessing the Work Ability Index of people with low vision

Abstract

Purpose: This investigation was aimed at clarifying the importance of the individual predictors and psychosocial working conditions in estimating the Work Ability Index (WAI) of people with impaired vision.

Materials and methods: This study utilized a quantitative cross-sectional design involving 450 respondents who had low vision with visual acuity in the range of 0.05–0.3 and 150 respondents with visual acuity in the range of 0.5–0.3 as a control group. The investigation was conducted using the standardized instruments Copenhagen Psychosocial Questionnaire (COPSOQ) and WAI.

Results: The average WAI score of the respondents with low vision was 30.1 ± 4.1 and that of the control group was 34.7 ± 3.4. Results indicated a positive COPSOQ/WAI correlation in the following domains: work importance, career prospects, rewards/feedback, role and the definition, quality of management, social support, job satisfaction, and health condition. Negative correlations were indicated in the following domains: quantitative, cognitive, and emotional demands; exhaustion; and stress.

Conclusions: The WAI of people with low vision is both positively and negatively impacted via different intensities associated with complex correlations, including gender; age; profession; stress propensity; quantitative, qualitative, and emotional demands of work; specificity of work, professional education; and the presence of stressors.

Keywords: individual predictors; psychosocial conditions; Work Ability Index; people with low vision

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1. INTRODUCTION

Vision impairment represents a clinical stage in the development of many eye diseases in which the terminal stage can evolve into a severe degree of visual impairment. This represents not only an ophthalmologic but also a sociological problem for the people affected [1]. Preserved eyesight is one of the main preconditions for the general population to function normally, which is the reason behind very visual impairment significantly affecting the quality of life, workability, and work performance of those affected regardless of their socio demographic characteristics and occupations [2]. Therefore, impaired vision is a significant contributor to morbidity and is strongly correlated with psychosocial and other conditions of the work environment. In addition, the visual degradation process significantly affects the workability, choice of profession, employment opportunities, work productivity, and assessment of the achieved results of people with low vision [3–5].

Workability is the ability of an individual to perform any type of work that can be valued, the ability of an organism to maintain an internal physiological balance during and after the completion of the activity, and the ability to quickly and thoroughly establish all the forms of physiological homeostasis that deviated from the normal values during work. In what is different from the case of those with preserved vision, whose workability is determined by the functioning of various organ systems and depends on the health, physical, and sensory ability of the organism, determining the workability of a person with low vision is a complex problem. Particularly, in addition to medical problems, many times, these impairments result in legal provisions, including driving restrictions. The Work Ability Index (WAI) is used universally to assess the workability of an individual [6,7].

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Healthy vision is fundamentally involved in various work processes and is particularly apparent in professions that require a high degree of visual preservation [8–10]. Few reports attempt to identify the predictors of the work environment that are most important and find out the extent to which each influences the preservation of workability (i.e., in the valorization of the WAI of people with low vision). This further emphasizes the need to define factors that affect workability, where by today's emphasis is placed not only on the health status of the people and the singularity of the job and the workload but also on the complex interrelated psychosocial conditions of the work environment that play the role of predictor and are crucially important in the assessment of the WAI.

Assessing workability determines the compatibility of the biological potentials and functions of an individual's organism with the requirements of a job and the characteristics of the work conditions. In everyday practice, workability can be expressed through the WAI [11–13]. Factors such as gender, age, profession, health status, individual and contextual personality traits, and psychosocial conditions at work affect the WAI scores [14–18].

Taking the large gaps in the literature regarding this topic as given, this investigation was aimed at clarifying the complex correlations, effects, and consequences of the influence of the mentioned factors on the workability of people with low vision from various professions. Moreover, we sought to determine the psychosocial conditions and other factors involved in the process of work that have the significance of a predictor, as well as the manner in which they affect the WAI of people with low vision. While researching the psychosocial factors, we analyzed the following parameters that affect the WAI: job requirements, work organization, singularity of the job and the workload, level of expertise and education, quality of

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management, general health status, individual and contextual personality traits, and management team. We have grouped these parameters into four dimensions: Job requirements; work organization and career prospect; relationships of employees with their superiors in the work process; and the effects/consequences of the psychosocial predictors in the work process.

The literature was searched thoroughly, and we use reference material and pre-knowledge, coordinated and supplemented by the results and findings obtained from several relevant studies [19,20].

Results from this investigation contribute to the illumination of the complex associations and individual, contextual, and psychosocial relationships to the workability of people with low vision who are involved in the work process, expressed through the WAI.

2. SUBJECTS AND METHODS

2.1. Study Design

This investigation utilized a cross-sectional study design with a quantitative analytical approach.

2.2. Subjects

Six hundred respondents were involved in the present study: 450 were people with low vision, such as those with glaucoma, diabetic retinopathy, and ophthalmic diseases with refractory anomalies and with preserved visual acuity of 0.05–0.03, and 150 people with preserved visual acuity of 0.5–0.3 were in the control group, according to the ICD-10 classification. The choice of respondents was based on inclusion and exclusion criteria

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related to age, the degree of visual impairment, and type of ophthalmic disease. Inclusion into the low vision group required low preservation of visual acuity, in the range of 0.05–0.3, and the control group required preservation of visual acuity of 0.5–0.3. The participant's age was required to be above 25 years old, with the condition that the low vision was caused by ophthalmological diseases such as glaucoma, diabetic retinopathy, and refractory anomalies.

2.3. Methods

Among the respondents involved in the study, visual acuity was determined by clinical ophthalmologic examination using appropriate ophthalmic equipment, which determined visual acuity at a distance and in proximity and the degree of visual preservation according to the ICD-10 classification.

The assessment of the impact of psychosocial conditions of the work process on the WAI of the respondents was conducted using the National Center for the Work Environment's Copenhagen Psychosocial Questionnaire (COPSOQ) and the WAI from the Finnish Institute of Occupational Health (FIOH). Both questionnaires have been designed and accepted in many European countries as standardized questionnaires in scientific research. All data were collected by trained examiners, and the respondent's involvement was voluntary. The anonymity and confidentiality of the answers were guaranteed, and the internal credibility and consistency of the data were tested with Cronbach's alpha coefficients.

COPSOQ has been designed and accepted as a standardized measuring instrument around the world and is used to study and follow the connection and impact of psychosocial factors of the work environment on workability. In the present

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investigation, we used a COPSOQ variant consisting of 23 domains (questions), 17 of which are dual, divided into five dimensions, and for each answer to set domains (questions), an adequate score of 1–5 points was calculated. The analysis of the scores obtained through the implementation of this test enabled us to derive basic understanding and arrive at conclusions regarding the connection and the impact of the psychological factors of the work environment on workability and risk assessment at work. Further more considering adequate measures and interventions for their elimination, this technique improves the work conditions and health preservation of employees and the prevention of morbid conditions [21,22].

The WAI questionnaire is a standardized questionnaire created by the FIOH and is used to assess the WAI in everyday life practice and scientific research. This test can identify the risk factors that affect workability; however, it can also be used as a selection method for the assessment of the results of intervention measures applied and as a program for detecting the influence of negative factors that affect the reduction of workability, with the aim of improving individual workability and the health of employees of various professions.

The WAI questionnaire consists of seven dimensions—related to self-assessed current workability; workability in relation to job requirements; number of current illnesses; assessment of inability to work due to illnesses; sick leave during the 12-month period; self-assessment of workability for 2 years; and psychic abilities - composed of one, two, or three domains (questions), and for each of the answers, 0–5 points are given. The total WAI cumulative score can range from 7 to 49 points. The assessment of the WAI was done by the calculated score of the points obtained, and if the total score of the answer was in the range of 7–27 points, the WAI score was

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considered to be poor, 28–36 points signify a moderate WAI score, 37–43 points indicated a good WAI score, and 44–49 points denoted an excellent WAI score [23,24].

2.4. Statistical Analysis

All data were processed using a descriptive and comparative statistics method, using statistical program SPSS 17.0. The attributive (qualitative) series were analyzed by determining the coefficients of ratio, proportion, and rate and were presented in absolute and relative numbers. Numerical (quantitative) series were analyzed by finding the measures of central tendency (average value; median, minimum, and maximum value; and interactive range) and dispersion measures (standard deviation and standard error).

The Pearson χ^2 test for homogeneity, Yates corrected test, and Fisher exact test were used to determine the difference between attributive dichotomous features.

Pearson (r) correlation coefficient was used to determine the association between numerical home group statistical series, and the Spearman correlation coefficient was used to determine the association between numerical non homogeneous statistical series.

In order to test the significance of the difference between the parameters analyzed, depending on the type and distribution of data, Student´s *t* test for two independent samples, ANOVA for multiple independent samples, and the nonparametric Mann–Whitney U test for independent samples were used.

A univariate regression analysis was used to assess the impact of psychosocial factors on the WAI, the coefficients of linear regression (B), as well as their 95% confidence intervals (CI), were shown, and the evaluation of the statistical significance of the coefficient value was completed using an unpaired *t* test. (Coefficients represent

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changes in the WAI score that are caused by increasing the score of independent variables for one measuring unit.)

A$p < 0.05$ was considered to be statistically significant.

3. RESULTS

All data collected from the respondents were analyzed according to socio demographic parameters such as gender, age, and profession. Of the respondents with low vision, 240 (53.3%) were female and 210 (46.7%) were male, and in the control group, 75 (50%) were male and 75 (50%) were female. One hundred and thirteen (25.1%) were 20–50 years old, and 337 (74.9%) were over 50 years old, and in the control group, 41 (27.3%) were aged 20–50 years old, and 109 (72.7%) were over 50 years of age. Of the respondents with low vision, 134 (29.8%) were office workers/administrators, 125 (27.8%) were laborers, 76 (16.9%) were farmers, and 115 (25.6%) were from other professions, and in the control group, the largest number, 47 (31.3%), were laborers, 39 (26%) were office workers/administrators, 21 (14%) were farmers, and 43 (28.7%) were from other professions.

The Work Ability Index of the low vision group and the control group subjects are shown in Tables 1 and 2.

Results from the comparison between the WAI of the respondents with low vision and the control group respondents indicate the presence of a statistically significant difference between the glaucoma/control group (Mann–Whitney U test: $Z = -7.079, p = 0.001$); diabetic retinopathy/control group (Mann–Whitney U test: $Z = -13.765, p = 0.001$); and refraction anomalies/control group (Mann–Whitney U test: $Z = -6.610, p = 0.001$). All low vision/control groups (Mann–Whitney U test:

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$Z = -11.217, p = 0.001$) were in favor of the better workability of control group respondents when compared with those with low vision, and so were the respondents from each low vision group individually. The specific dimensions of the WAI index regarding "Estimated work impairment due to disease" are shown in Table 2. There was a significant difference in WAI scores between glaucoma patients and those with diabetic retinopathy (Mann–Whitney U test: $Z = -7.493, p = 0.001$) and in WAI scores between patients with refractory anomalies and those with diabetic retinopathy (Mann–Whitney U test: $Z = -7.395, p = 0.001$) in favor of better working abilities of glaucoma patients and those with refractory anomalies than those with diabetic retinopathy; however, there was no statistically significant difference in the WAI scores of glaucoma patients in relation to refractory anomalies (Mann–Whitney U test: $Z = -0.322, p = 0.747$).

The low vision and the control group respondents were divided into two subgroups: those aged 20–50 years old and those over 50 years of age. The WAI scores of the respondents in relation to age are shown in Table 3.

The correlation of WAI scores between respondents with low vision and the control group aged 20–50 years old and those over 50 indicates the presence of a statistically significant difference in favor of better workability of the control group than the respondents with low vision (difference between two means − two-sided:$p = 0.001$).

The correlation with parameters of the WAI score/age of the respondents with low vision (Spearman rank order correlation: $R = -0.121, p < 0.05$) and the control group respondents (Spearman rank order correlation: $R = -0.172, p < 0.05$) of both genders, aged 20–50/over 50, indicates that there was a statistically significant linear negative correlation and that as age increased, the workability of the respondents decreased in a linear fashion.

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The WAI scores of the respondents in relation to gender are shown in Table 4.

The analysis of the results suggests the presence of a statistically significant difference between male and female respondents, in favor of better workability of male respondents than female ones (Mann–Whitney U test: $Z = -2.191$, $p = 0.02$).

There was a significant difference between the control group male respondents, who had aWAI score of 35.1 ± 3.5, and male respondents with low vision, who had a WAI score of 30.39 ± 4.1, in favor of better workability of the control group male respondents (Difference between two means − two-sided: $p = 0.008$).

There was a significant difference between the control group female respondents whose WAI score was 34.4 ± 3.3 and the female respondents with low vision, who had a WAI score of 29.7 ± 4.1, in favor of improved workability of the control group female respondents (Difference between two means − two-sided: $p = 0.001$).

The correlation parameters of the WAI score of the respondents with low vision (Spearman rank order correlation: $R = -0.103$, $p > 0.05$) and those of the respondents from the control group (Spearman rank order correlation: $R = -0.111$, $p > 0.05$) of both genders, aged 20–50/over 50, suggests an insignificant negative linear correlation in favor of reduced workability of the low vision and control group female respondents, which may reduce with age.

We also tested the WAI scores on all respondents with low vision and control group respondents according to occupation (Tables 5 and 6).

The variance analysis (one-way ANOVA) revealed a significant difference in professions between respondents with low vision and control group respondents in all correlated combinations of professions, with higher WAI scores of control group respondents.

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WAI scores are categorized as follows: poor, moderate, good, and excellent, and this categorization is shown in Table 7. A WAI score of7–27 points indicates poor workability; 28–36 points denotes moderate workability, 37–43 points signifies good workability, and 44–49 points suggests excellent workability. A correlation analysis of the WAI of the low vision/control group indicates that there was a significant difference in glaucoma/diabetic retinopathy patients(Fisher–Fridman–Halton test:$p = 0.001$) in favor of better working abilities of patients with glaucoma; glaucoma/refractory anomaly (Pearson $\chi^2 = 7.81$, df = 2, p = 0.002) in favor of better working abilities of patients with refractory anomalies; diabetic retinopathy/refraction anomalies (Fisher–Fridman–Halton test :$p = 0.001$) in favor of better working abilities of patients with refraction anomalies; and in low vision/control groups (Fisher–Fridman–Haltontest:$p = 0.001$) in favor of better performance of control group respondents.

Results of the psychosocial factors of the work environment from the WAI of the respondents are shown in Tables 8 and 9.

With the univariate regression analysis from the dimension of WORK DEMANDS of low vision and control group respondents, it was observed that an increase in COPSOQ score for one was associated with decline on the WAI, with a significant difference in the domains of quantitative, cognitive, and emotional demands ($p < 0.001$).

In the dimension ORGANIZATION OF WORK AND CAREER PROSPECTS, it was observed that in all low vision and control group respondents, an increase in COPSOQ score for one was related to an increase in the Work Ability Index with statistically significant difference in the work importance and career prospects domains ($p < 0.001$), where as in the work duties and employee's influence at work domains, an

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increase in COPSOQ score for one also positively affected workability but without any statistical significance ($p > 0.05$).

In the RELATIONSHIPS BETWEEN EMPLOYEES AND SUPERVISORS IN THE WORK PROCESS dimension, it was observed that in all low vision and control group respondents, an increase in COPSOQ score for one was associated with an increase in the Work Ability Index with a statistically significant difference in rewards/feedback, role and definition, social support, and quality management ($p < 0.001$). In addition, in the predictability, employee–supervisor relationships, and fairness and reputation domains, although there was a positive correlation, a statistically significant difference was not observed.

In the EFFECTS/CONSEQUENCES dimension, univariate regression analysis revealed that in all low vision and control group respondents, an increase in COPSOQ score for one was associated with an increase in the Work Ability Index with a significant difference in the job satisfaction and health status domains ($p < 0.001$). An increase in COPSOQ score for one was associated with a decline in the WAI with a statistically significant difference in the exhaustion and stress domains ($p < 0.001$). In the work/private life conflict, the increase in COPSOQ score for one was associated with a reduction in the WAI score. However, this association failed to reach statistical significance.

The results of the exposure of respondents to mobbing are shown in Table10. For 7.16% of the respondents, the mobbing was carried out in the form of intimidation; hence, their recording indicates that mobbing is an integral part of negative psychosocial influence in the work process. There was no statistically significant difference between the number of threats made to low vision and control group respondents (Pearson $\chi^2 = 0.523$, df = 1, p = 0.469); however, with both groups of

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respondents, the risk of exposure to this and other types of mobbing and discrimination exists.

4. DISCUSSION

Recent assessments of the workability of different categories of people depending on the occupation indicate that the Work Ability Index can be used as an objective means to evaluate workability and identify risk factors of reduced workability. Moreover, it is suitable for assessing the effects of the applied intervention program and suggesting measures for the elimination of unfavorable and risky work environment factors that adversely affect workability. The WAI is also suitable as a valid identifier of applied preventive programs for the promotion of occupational health [12,18,23,25].

Vision impairment is an ophthalmologic condition that has multi-causal etiology, has complex pathogenesis, and can evolve into severe forms of damage and definite loss of vision, where upon it becomes a handicap that affects not only the quality of life but also the ability to work [1].

People with vision impairment, in addition to other adverse effects, are exposed to continuous stress, unfavorable psychosocial conditions in all organizational stages of the work process, frequent mobbing, and discrimination during work, all of which reflect on their workability and productivity [16,26].

In 34 low vision and 9 control group respondents (7.16%), mobbing was conducted in the form of intimidation or another type of discrimination during work. Khoo and associates [27] stated that during working hours, people with disabilities were very often under constant stress not only because of the particularities of their work but also because of the tendency to be very often discriminated against and marginalized by colleagues and the surroundings in general. Sanders [28] and Alavinia [29], in similar

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investigations, emphasized the subjective, emotional, and contextual characteristics of the personality of the disabled as an additional negative factor in their self-assessment of health and productive potentials. Both investigations are in accordance with our findings. Several previous investigations conducted with people from different professions indicated that workability, in parallel with the function of eyesight, was affected by other factors. However, they do not fully clarify the complex relationships of specific individual characteristics and the intensity of perceptions of different relationships and actions, particularly psychosocial factors in the work process[30].

The WAI scores of the low vision group, when compared with the scores of the control group respondents, were significantly different from the low vision/control groups in favor of the better workability of control group respondents than those with low vision.

Comparisons of the WAI results with similar parallel scientific studies suggest that all respondents with low vision involved in our research obtained a significantly lower average WAI score than that obtained by respondents with similar socio demographic characteristics but with preserved vision. Lavasani[31], in a study carried out on 275 employees with various disabilities in Malaysia, claimed that 29.5% had poor workability, 35.3% had moderate workability, 28.7% had good workability, and 18% had excellent workability.

We observed that respondents aged 20–50 years old have better workability when compared with subjects over the age of 50 years, with a statistically significant difference, including both low vision and control group respondents. In previous studies on different aspects of age and workability, several authors refer to the existence of a negative correlation between workability and age [32–35].

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There was a statistically significant difference between the low vision and control group male and female respondents in favor of the better workability of male respondents when compared with female ones. The Work Ability Index of male and female participants of the low vision group and the control group, with age, showed reduced scores, which, for male and female respondents over the age of 50 years, was statistically significantly lower when compared with the WAI scores of both genders younger than 50 years old. Van den Berg [36] in a study of multifactorial etiology of the influence of the WAI scores on professional employees from different occupations suggested that male respondents have a higher WAI score for two index points (4%) when compared with the female ones. With regard to age, the WAI scores for employees over 40 years of age were considered to decrease by four index points (7.3%) when compared with younger employees. In his study, it is also evident that each of the psychosocial factors regarding gender have different unique variables of influence on workability.

Both Rotenberg [15] and Claudie Fernandes [37] reported that with both men and women, there were differences in the WAI scores in relation to gender in that the WAI scores in female employees were lower in relation to men and that the vulnerable period in which there was a significant reduction in WAI scores of women was uniquely expressed in the age groups of 25–34, 40–44, and 60–64 years old.

Using a Tukey *post hoc* analysis, we assessed the average WAI scores in relation to the level of education and professional orientation of all the 450 respondents with low vision and found a statistically significant difference in relation to the laborers/office workers ($r = 0.003$) in favor of the better workability of office workers; laborers/farmers ($r = 0.005$) in favor of better workability of laborers; laborers/others ($r = 0.004$) in favor of the better workability of respondents from other professions; office workers/farmers

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(r = 0.001) in favor of better workability of office workers; farmers/others (r = 0.003) in favor of the better workability of respondents from other professions. Lavasani[31] aimed at measuring the workability of employees with a disability and reported that WAI for employees with primary education was 28.92 ± 8.36; with secondary education, 30.99 ± 7.77; and with higher education, 41.11 ± 4.47. In similar research focused on the workability of various categories of medical and nursing staff in hospitals, Ehsanollah found that the WAI score was 38.25 ± 4.4 and that there were WAI variations regarding the level of education, qualifications, and sub-professional specifications [38].

In a survey conducted with employees in commercial activities,Van den Berg [36] noted that the average WAI score was41.1 ± 5.1 and was greater than the average WAI of employees in mentally demanding jobs (reported average of39 ±4.2 as conducted by Tuomi [25]). Sjögren–Ronka [39] also emphasized the negative impact of physical demands and stress factors on the workability of office workers employed in various positions.

The influence of the psychosocial conditions of work environments on workability in our study was assessed from 19 different aspects (domains), with univariate regression analysis. It was observed that with low vision and control group respondents, an increase in COPSOQ score for one was associated with an increase in the WAI score with statistically significant difference ($p < 0.001$) in the domains of work importance, career prospects, rewards/feedback, role and definition, quality of management, social support, job satisfaction, and health status, which indicates that improved conditions related to these domains favorably influence an increase in workability and that it is necessary to try to maintain or improve their existing gradient.

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In the domains of quantitative demands, cognitive demands, emotional demands, and exhaustion and stress, an increase in COPSOQ score of one was associated with a decline in the WAI score with a statistically significant difference ($p < 0.001$). This result implies that people with low vision, particularly those with severe visual impairment, should take preventive measures and avail of programs that will enable its elimination. The measures taken will enable the improvement of workability and health protection of people with low vision.

There was no significant impact of psychosocial factors on the WAI score in the domains of workplace duties, predictability, employee–superior relationships, fairness, reputation, and work/family conflicts, although, depending on the domain, there was a negative/positive impact of the psychosocial conditions of the work environment on the WAI.

In the interim, there are very few reference reports that address the problem of workability of people with impaired vision. Moreover, detecting crucial psychosocial determinants that play the role of predictor of workability and identifying the gradient at which they have a positive or negative influence was challenging for us to try and contribute to the illumination of this complex and exact problem. Our research results were compared with the results of previous scientific studies conducted in different countries of the world and relating to the influence of psychosocial conditions on the WAI of other categories of healthy and disabled people from different professions. Van den Berg [36] assessed the impact of psychosocial factors on employee health and workability and found that workability was strongly associated with stress, the organization of work processes, physical demands of work, the individual mental perception of stress factors, and the possibility of their balancing by employees.

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Edwards et al. [40], in a study on the effect of psychosocial factors on the workability of nurses, emphasized the importance of factors such as physical overload, employee/medical staff relationships, and bureaucratic management in performing work tasks. Borritz[41] also showed the effect of physical exhaustion and overload on individual potentials and managerial relationships in the work process.

5. CONCLUSIONS

Results from this investigation suggest the complicated influence of the psychosocial aspects of work environments on the WAI of low vision and control group respondents.

Vision impairment in itself does not have a decisive importance in determining or assessing the Work Ability Index. However, there exists a significant correlation with the effects of psychosocial conditions of the work environment, such as the quantitative, qualitative, and emotional demands of work; specificity of the type of profession; vocational training and education; exposure and intensity of stressors at work; general health condition and basic illnesses; and individual personal traits; among other circumstances. Crucially, these are not merely related to the level of vision impairment, suggesting that other factors are perhaps at play.

Timely consideration and detection of the psychosocial factors of the work environment, which have a negative impact on the WAI, will enable adequate preventive measures to be taken for their elimination and will contribute to not only the improvement of workability but also to preserving health over the long term and preventing disease from afflicting anyone involved in the work process, particularly those with vision impairment.

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Table 1. WAI scores of the respondents

Disease	No	Average WAI Score (Mean)	Standard Deviation	95% CI for Mean		Minimum	Maximum
				Lower Bound	Upper Bound		
Glaucoma	150	31.2	3.9	30.6	31.9	25.9	37.2
Diabetic retinopathy	150	27.5	2.5	27.1	27.9	23.7	31.8
Refraction anomalies	150	31.4	4.3	30.7	32.1	25.3	37.6
Low vision group	450	30.1	4.1	29.7	30.4	23.7	37.6
Control group	150	34.7	3.4	34.2	35.3	28.6	40.6

Note: WAI = Work Ability Index; CI = confidence intervals.

Table 2. WAI scores of the low vision respondents based on diseases

Mann–Whitney U test	Disease		
	GL/ RE	GL/ RA	RE/ RA
Z	−7.493)	−0.322)	−7.395)
Asymp. Sig. (2-tailed)	0.001*	0.747	0.001*

*significance of $p < 0.05$

Note: GL = glaucoma; RE = diabetic retinopathy; RA = refraction anomalies.

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Table 3. WAI scores of respondents based on age

Age	No.	Average WAI Score (Mean)	Standard Deviation	95% CI for Mean		p
				Lower Bound	Upper Bound	
Glaucoma						
20–50	37	31.75	3.78	30.48	33.01	**Mann–WhitneyUtest:**
>50	113	31.09	3.96	30.35	31.83	**Z=1.875 *p*=0.049***
Diabetic retinopathy						
20–50	38	28.19	2.44	27.38	28.99	**Mann–WhitneyUtest:**
>50	112	27.22	2.55	26.75	27.70	**Z=2.954 *p*=0.003***
Refraction anomalies						
20–50	38	32.03	4.15	30.67	33.39	**Mann–WhitneyUtest:**
>50	112	31.18	4.29	30.38	31.99	**Z=2.455 *p*=0.014***
Low vision group						
20–50	113	30.64	3.92	29.91	31.37	**Mann–WhitneyUtest:**
>50	337	29.83	4.11	29.39	30.27	**Z=2.561 *p*=0.010***
Control group						
20–50	41	35.55	3.22	34.53	36.57	**Mann–WhitneyUtest:**
>50	109	34.43	3.43	33.78	35.08	**Z=2.098 *p*=0.036***

*significance of $p < 0.05$

Note: WAI = Work Ability Index; CI = confidence intervals.

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Table 4. WAI scores of respondents based on gender

Diseases	No.	Average WAI Score (Mean)	Standard Deviation	95% CI for Mean		p
				Lower Bound	Upper Bound	
Glaucoma						
male	81	31.47	3.89	30.61	32.33	**Mann–WhitneyUtest: Z=−1.722 *p*=0.085**
female	69	30.99	3.97	30.04	31.95	
Diabetic retinopathy						
male	63	27.68	2.56	27.03	28.32	**Mann–WhitneyUtest: Z=−1.339 *p*=0.181**
female	87	27.32	2.54	26.78	27.86	
Refraction anomalies						
male	66	31.67	4.24	30.62	32.71	**Mann–WhitneyUtest: Z=−1.467 *p*=0.142**
female	84	31.19	4.29	30.25	32.12	
Low vision group						
male	210	30.39	4.06	29.84	30.94	**Mann–WhitneyUtest: Z=−2.191 *p*=0.028***
female	240	29.73	4.07	29.21	30.25	
Control group						
male	75	35.05	3.52	34.24	35.86	**Mann–WhitneyUtest: Z=−1.355 *p*=0.175**
female	75	34.42	3.27	33.67	35.17	

*significance of $p < 0.05$

Note: WAI = Work Ability Index; CI = confidence intervals.

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Table 5. WAI scores of respondents with low vision based on profession

Profession	Average WAI Score (Mean)	No.	Standard Deviation	Minimum	Maximum	p
Laborer	29.4	125	3.2	24.1	29.4	**One-Way ANOVA: F=17.854 df=3; *p* =0.001***
Office worker	31. 1	134	4.8	23.8	31.1	
Farmer	27.6	76	2.8	23.7	27.6	
Other	31.1	115	3.95	24.2	31.1	

* significance of $p < 0.05$

Note: WAI = Work Ability Index.

Table 6. WAI scores of control group respondents based on profession

Profession	Average WAI Score (Mean)	No.	Standard Deviation	Minimum	Maximum	p
Laborer	34.9	47	3.6	29.1	40.2	**One-Way ANOVA: F=7.858 df=3; *p*=0.007***
Office worker	36.3	39	3.2	28.6	40.6	
Farmer	32.2	21	2.1	29.3	36.1	
Other	34.2	43	3.2	28.9	39.2	

*significance of $p < 0.05$

Note: WAI = Work Ability Index.

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Table 7. Attributive values of WAI of respondents in the study

Disease		WAI			Total
		Poor	Moderate	Good	
Glaucoma	**Number**	42	102	6	150
		28%	68%	4%	
Diabetic retinopathy	**Number**	76	74	0	150
		50.7%	49.3%	0%	
Refraction anomalies	**Number**	43	88	19	150
		28.7%	58.7%	12.6%	
Low vision group	**Number**	161	264	25	450
		35.8%	58.7%	5.5%	
Control group	**Number**	16	84	50	150
		10.7%	56%	33.3%	

* significance of $p < 0.05$

Fisher–Fridman–Halton test: $p = 0.001$* glaucoma/ diabetic retinopathy
Pearson $\chi^2 = 7.81$, df = 2, $p = 0.002$* glaucoma/ refraction anomalies
Fisher–Fridman–Halton test: $p = 0.001$* d.retinopathy/refractionanomalies
Fisher–Fridman–Halton test: $p = 0.001$* low vision/control group

Note: WAI = Work Ability Index.

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Table 8. Connection of COPSOQ scores to WAI scores of respondents with low vision

COPSOQ DOMAINS	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95% CI for B	
	B	SEM	Beta			Lower Bound	Upper Bound
Quantitative demands	−1.558	0.055	−0.803	−28.529	0.001	−1.666	−1.451
Cognitive demands	−1.627	0.055	−0.816	−29.835	0.001	−1.734	−1.520
Emotional demands	−1.595	0.061	−0.775	−25.955	0.001	−1.716	−1.474
Employee's influence at work	1.388	0.064	0.661	18.645	$p>0.05$	1.068	1.319
Career prospects	1.326	0.056	0.743	23.515	0.001	1.215	1.437
Work importance	1.193	0.064	0.661	18.645	0.001	1.068	1.319
Work duties	1.193	0.055	0.769	25.459	$p>0.05$	1.281	1.495
Predictability	1.279	0.047	0.790	27.319	$p>0.05$	1.187	1.371
Rewards/feedback	1.388	0.055	0.769	25.459	0.001	1.281	1.495
Role and definition	1.622	0.065	0.762	24.911	0.001	1.750	1.494
Quality management	1.916	0.078	0.757	24.535	0.001	1.762	2.069
Social support	1.588	0.059	0.787	26.959	0.001	1.472	1.704
Relationship employee – supervisor	1.197	0.062	0.671	19.159	$p>0.05$	1.074	1.320

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Fairness and reputation	1.254	0.049	0.768	25.375	$p>0.05$	1.157	1.351
Job satisfaction	3.026	0.122	0.761	24.839	0.001	2.787	3.266
Health condition	2.849	0.081	0.856	35.008	0.001	2.689	3.009
Exhaustion	−1.595	0.057	−0.796	−27.815	0.001	−1.708	−1.482
Stress	−1.598	0.056	−0.802	−28.389	0.001	−1.709	−1.487
Job/private life conflict	−1.193	0.064	0.661	18.645	$p>0.05$	1.068	1.319

a. Dependent Variable: WAI score *significance of $*p < 0.05$

Note: B = coefficient of linear regression; SEM = standard error of the mean; t = *t* test; Sig. = statistical significance.

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Table 9. Connection of COPSOQ scores to WAI scores of control group respondents

COPSOQ DOMAINS	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95% CI for B	
	B	SEM	Beta			Lower Bound	Upper Bound
Quantitative demands	−1.602	0.064	−0.901	−25.212	0.001	−1.727	−1.476
Cognitive demands	−1.672	0.071	−0.887	−23.387	0.001	−1.813	−1.531
Emotional demands	−1.579	0.083	−0.843	−19.083	0.001	−1.742	−1.415
Employees influence at work	1.305	0.047	0.915	27.671	$p>0.05$	1.212	1.398
Career Prospects	1.282	0.048	0.910	26.766	0.001	1.187	1.377
Work importance	1.305	0.047	0.915	27.671	0.001	1.212	1.398
Work duties	1.313	0.046	0.921	28.691	$p>0.05$	1.223	1.404
Predictability	1.451	0.047	0.931	30.931	$p>0.05$	1.358	1.543
Rewards/feedback	1.313	0.046	0.921	28.691	0.001	1.223	1.404
Role and definition	1.761	0.066	0.910	26.679	0.001	1.892	1.631
Quality management	1.958	0.071	0.916	27.733	0.001	1.818	2.097
Social support	1.451	0.047	0.931	30.931	0.001	1.358	1.543
Relationship employee–supervisor	1.240	0.046	0.912	27.092	$p>0.05$	1.149	1.330
Fairness and	1.243	0.040	0.932	31.229	$p>0.05$	1.164	1.321

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reputation							
Job satisfaction	2.938	0.096	0.929	30.642	0.001	2.748	3.127
Health condition	2.627	0.103	0.902	25.394	0.001	2.423	2.831
Exhaustion	−1.741	0.066	−0.908	−26.403	0.001	−1.871	−1.610
Stress	−1.579	0.083	−0.843	−19.083	0.001	−1.742	−1.415
Job/private life conflict	−1.305	0.047	0.915	27.671	$p>0.05$	1.212	1.398

a. Dependent Variable: WAI score *significance of*$p < 0.05$

Note: B = coefficient of linear regression; SEM = standard error of the mean; t = *t* test; Sig. = statistical significance.

Table10. Exposure to intimidation

Respondents		**Exposure to intimidation**		**Total**
		Yes	**No**	
Low vision	**Number**	34	415	450
	%	7.55%	92.22%	
Control	**Number**	9	141	150
	%	6%	94%	
Total	**Number**	43	556	600
	%	7.16%	92.67%	100%

Pearson χ^2= 0.523,df = 1, $p = 0.469$* significance of *$p < 0.05$

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